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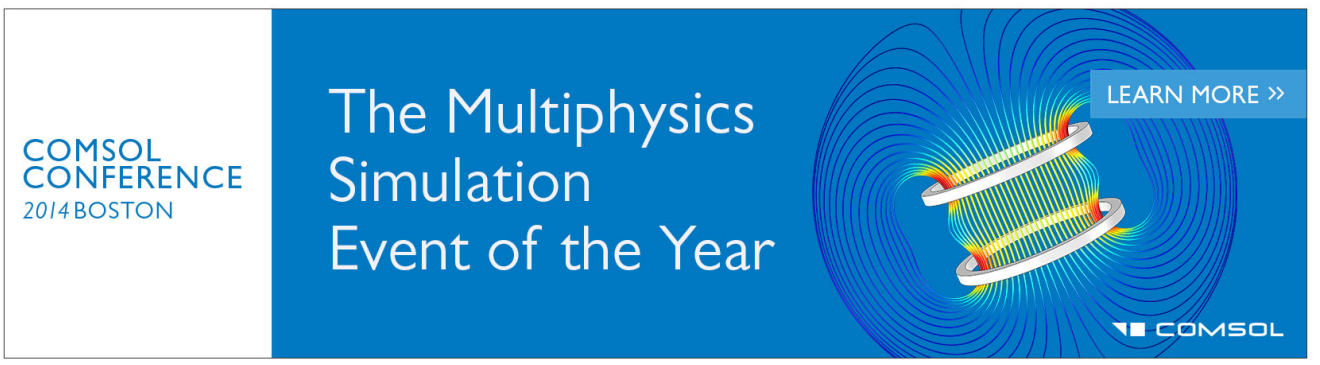
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Damping of lower hybrid waves in large spectral gap configurations[a)]

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Extensive experimental data support reliable power deposition and current drive by lower-hybrid (LH) waves in conditions where a large spectral gap exists between the nominal parallel index of refraction prescribed by the antenna characteristics and phasing, and that required for significant Landau damping to take place. We argue that only a significant modification of the initial spectrum at the plasma edge could explain experimental observations. Based on this assumption, a new prescription for reliable simulations of LH current drive using ray-tracing and Fokker-Planck modelling is proposed. A remarkable agreement between experimental observations in the Tore Supra tokamak and simulations is obtained for relevant parametric scans, including electron density and LH waveguide phasing. In an effort to investigate the possible role of fluctuations, it is shown that the spectral gap can be bridged dynamically in the presence of a fluctuating LH spectrum. © *2014 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4894749]

I. INTRODUCTION

Lower hybrid (LH) waves can exchange energy and parallel momentum with electrons via Landau damping, following the resonance condition $v_{||} = c/n_{||}$ relating the parallel velocity $v_{||}$ of the electron to the wave parallel index of refraction $n_{||}$.[1] Detailed experimental analysis of LH current drive (CD) efficiency and fast electron bremsstrahlung measurements indicate the existence of a large population of electrons with a velocity in the vicinity of $v_{||0} = c/n_{||0}$ corresponding to the initial index of refraction $n_{||0}$ prescribed by the antenna.[2] However, in many past and present day LHCD experiments, the resonant velocity $v_{||0}$ is positioned far out in the suprathermal region, where the number of electrons should in principle be too small for any significant wave damping to take place. Indeed, strong wave damping on a Maxwellian distribution with temperature T requires the wave damping to be no larger than four times the thermal velocity $v_T = \sqrt{kT/m}$. This paradox is commonly referred to as the *spectral gap problem*.[3]

It is generally assumed that a tail of fast electrons is somewhat pulled from the Maxwellian distribution such that LH wave spectral components with $n_{||} \sim n_{||0}$ can be strongly damped. A variety of mechanisms have been suggested to explain how this fast electron tail is formed.[3] Among these, a majority rely on some modification of the LH spectrum between the grill and the damping region such that a fraction of the power is carried by spectral components $n_{||0} \lesssim n_{||} \lesssim n_{||\mathrm{LD}}$ where $n_{||\mathrm{LD}} \equiv c/(4v_T)$ characterizes strong Landau damping on a Maxwellian distribution.

The parallel index of refraction $n_{||}$ of LH waves propagating in a torus varies as a consequence of the poloidal non-uniformity of the magnetic field. The $n_{||}$ evolution is naturally described by the ray-tracing formalism,[4,5] which is valid for the description of radio-frequency waves in plasmas as long as the WKB and well-collimated beam approximations are valid, which can be expressed by the inequality $\lambda \ll d \ll L$, where λ is the wavelength, d is the beam size, and L is the equilibrium non-uniformity scalelength. The combination of ray-tracing (RT) calculations, which determine the $n_{||}$ evolution, and Fokker-Planck (FP) calculations, which describe the formation of a tail of fast electrons, has consequently become a standard for solving the spectral gap problem and simulate LHCD experiments.[6–8]

While RTFP modelling successfully compares with experiments in the presence of a narrow spectral gap, it fails with most current day experiments where the spectral gap is large, as found for instance in high-density conditions.[9] In such weak damping regimes, LH rays undergo many passes across the plasma before any power deposition occurs. LHCD calculations become very sensitive to equilibrium variations and initial conditions as the wave propagation becomes stochastic beyond a few reflections on the plasma edge.[10,11] Statistical methods were developed to bridge the spectral gap by filling the phase space with many rays.[12] While this approach did not successfully reproduce experimental results, it showed the existence of a LH propagation domain within which $n_{||}$ variations due to toroidicity are bounded. Recently, full-wave modeling in toroidal plasmas has been applied to LH wave propagation.[13,14] Interestingly, this approach showed similar features to ray-tracing modeling while offering a better description of the wave propagation near the plasma edge. Yet, it does not compare well with experiments in large spectral gap configurations.

Moreover, LH waves were successfully used to drive the plasma current for several minutes in the large aspect ratio tokamak TRIAM-1M where $n_{||}$ variations due to toroidicity are negligible.[15] For such near-cylindrical plasma in weak damping regime, both RT and full-wave description completely fail at bridging the spectral gap.[16] It is thus legitimate to search for alternative mechanisms. Recently, effects of plasma edge physics on the LH spectrum have been

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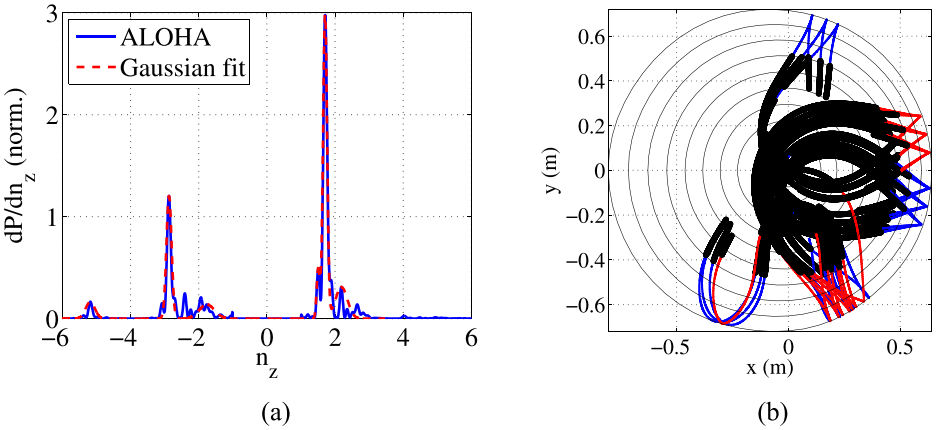


FIG. 1. (a) ALOHA LH spectrum and six main resulting Gaussian lobes. (b) Poloidal projection of the LH wave propagation represented by 36 rays combining six spectral lobes with six poloidal locations corresponding to waveguide arrays of the LH antenna.

extensively investigated, including parametric decay instabilities (PDI),[17–19] ponderomotive effects,[20] collisional damping,[21] wave scattering,[22,23] drift wave turbulence,[24] and blobs.[25]

From the universality and reproducibility of LHCD experimental trends (as a function of density, $n_{||}$, plasma current, etc.), we speculate that the spectral gap is filled at the plasma edge such that the power is rapidly absorbed in one or very few passes.[26] Rather than focusing on one particular mechanism to bridge the spectral gap, we propose to answer the following question: how much would an initial spectrum need to be modified to account for experimental observations? By constructing a parameterized spectrum at the last closed magnetic flux-surface (LCFS), we show that a significant modification of the spectrum at the edge is required to fill the spectral gap. Conversely, we propose a method to extend the use of RTFP modelling to the weak damping regime. Using this approach, essential parametric dependencies observed during LHCD in Tore Supra are reproduced by simulations for the first time. In an effort to investigate further the possible mechanisms at play, we show that our new method can be equivalent to the calculation of LHCD in the presence of a fluctuating spectrum.

Section II describes the RTFP modelling scheme used for Tore Supra LHCD simulations. The spectral gap problem is investigated in Sec. III, where a new prescription for RTFP modelling in large spectral gap configurations is proposed. In Sec. IV, it is shown that this method, which correctly reproduces experimental observations in Tore Supra, is equivalent to standard RTFP modelling in the presence of a fluctuating spectrum. This comparison provides a physical interpretation to the prescription in Sec. III while guiding future work towards an *ab-initio* investigation of the spectral gap problem.

II. RAY-TRACING FOKKER-PLANCK CALCULATIONS

The interpretative simulation of Tore Supra LHCD discharges uses a suite of codes that have been validated both separately and together.[2] The evolution of the discharge is calculated by an equilibrium/transport code such as METIS or CRONOS[27] related to the Tore Supra database. The LH wave propagation is determined by the ray-tracing code C3PO,[28] which uses the spectrum from the coupling code ALOHA[29] as illustrated in Fig. 1. The LH spectrum calculated by the coupling code ALOHA is typically processed into two to six Gaussian lobes $(n_{||i0}, \Delta n_{||i0}, P_{i0})$. The spectral location $n_{||i0}$ of the main lobes is selected by order of decreasing power, while the Gaussian width $\Delta n_{||i0}$ and the power associated with each lobe P_{i0} are adjusted such as to conserve the intensity of the main lobes, the total wave power, and the directivity.

While it is legitimate to assume that the ray trajectories are not significantly affected by the presence of fast electrons, it is necessary to calculate the wave damping and the electron distribution function self-consistently.[8] Consequently, an iterative scheme is used to solve the Fokker-Planck equation with the code LUKE.[30] As detailed in the Appendix, an essential aspect of this calculation for the spectral gap problem is that a convergence on the power flow along each ray must be ensured for a proper build-up of the quasilinear plateau.[31]

Useful moments of the electron distribution function include the driven current and LH power deposition profiles, which can be used to refine the equilibrium/transport calculation in a more global convergence scheme. In non-inductive discharges, the driven current can be compared to the plasma current to assess the quality of the simulation. In Tore Supra (as shown in Fig. 2) and other tokamaks, tomographic systems of hard X-ray (HXR) detectors were installed specifically to observe the fast electron dynamics by measuring the fast electron bremsstrahlung (FEB), with a combined resolution in space, time, and energy. The synthetic diagnostic R5X2 was developed to calculate the line-integrated FEB using the electron distribution from LUKE for a direct comparison between simulations and experimental observations.[32] One of the most remarkable, universal features of FEB measurements during LHCD experiments is that no signature of stochastic wave propagation effects is observed.[2,10,33,34] To the contrary, clear, smooth, and reproducible parametric dependences are observed, as a function of density or antenna phasing as illustrated in Fig. 5 for instance. Since the FEB and LH-driven current are generated by the same population of fast electrons, the driven current must also exhibit a deterministic behavior. The relative variations of the driven current in time must be of a similar amplitude to the variations in equilibrium and LH wave parameters.

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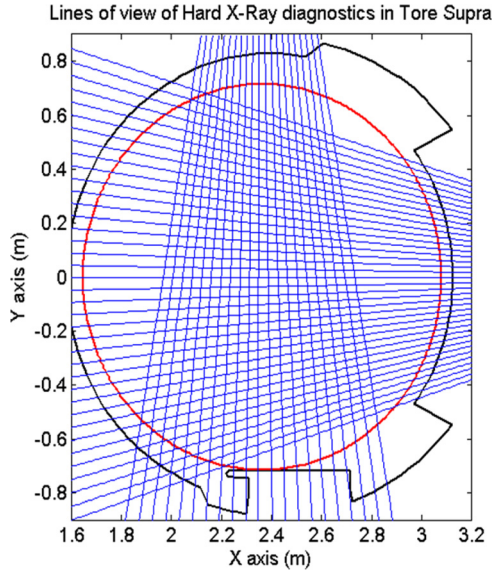


FIG. 2. Lines of sight illustrating the HXR tomographic system in the Tore Supra tokamak.

The modelling scheme is first applied to a Tore Supra discharge #45525 at $t = 27$ s (plasma current $I_p = 0.5$ MA, toroidal field on axis $B_t = 2.7$ T, volume-averaged density $\langle n \rangle = 1.3 \times 10^{19}\,\mathrm{m}^{-3}$, central temperature $T_{e0} = 5.7$ keV, LH power $P_{\mathrm{LH}} = 2.5$ MW). In the presence of a narrow to moderate spectral gap (main lobe $n_{\|0} = 1.7 - 1.8$; $n_{\|\mathrm{LD}} = 2.7$), the LH wave propagation shows no effect of stochasticity, and trajectories corresponding to three successive times are very similar as seen in Figs. 3(a)–3(c). Consequently, LHCD calculations are robust with respect to small variations in the equilibrium, and properly account for variation in the LH spectrum such as the change from $n_{\|0} = 1.72$ to $n_{\|0} = 1.82$ due to a modification of antenna phasing at $t = 20$ s (Fig. 4(a)).

When used in the presence of a large spectral gap, however, the modelling scheme yields very different results as illustrated by the Tore Supra discharge #45155 at $t = 9$ s ($I_p = 1.0$ MA, $B_t = 3.8$ T, $\langle n \rangle = 3.8 \times 10^{19}\mathrm{m}^{-3}$, $T_{e0} = 2.4$ keV, LH power $P_{\mathrm{LH}} = 2.1$ MW). Since the large spectral gap (main lobe $n_{\|0} = 1.8$; $n_{\|\mathrm{LD}} = 4.2$) is no longer easily bridged by toroidal upshift, LH rays are in a multi-pass regime for which a stochastic behavior clearly takes place as shown in Figs. 3(d)–3(f). The wave propagation, hence LHCD calculations, are very sensitive to small variations in the equilibrium, and results become erratic (Fig. 4(b)). These large variations in the driven current, which can exceed 50%, are not physical as the corresponding variations found in the simulated integrated fast electron bremsstrahlung signal are not observed in experimental measurements (see Fig. 5(a)). Clearly, RTFP modelling fails in this weak damping regime, as confirmed by simulations using a similar scheme in the Alcator C-Mod tokamak.[10]

III. THE SPECTRAL TAIL MODEL

The absence of stochastic wave propagation effects in experimental LHCD observations suggests that the LH spectrum may be significantly modified by plasma edge physics, thereby bridging the spectral gap such that the LH wave power is absorbed after a short path. To test this assumption, we construct a spectrum at the LCFS using a distribution of adjacent lobes to create a spectral tail from the initial value $n_{\|0}$ (prescribed by antenna phasing) to a maximum value $n_{\|\mathrm{max}}$. The spectral width associated with the Gaussian lobes

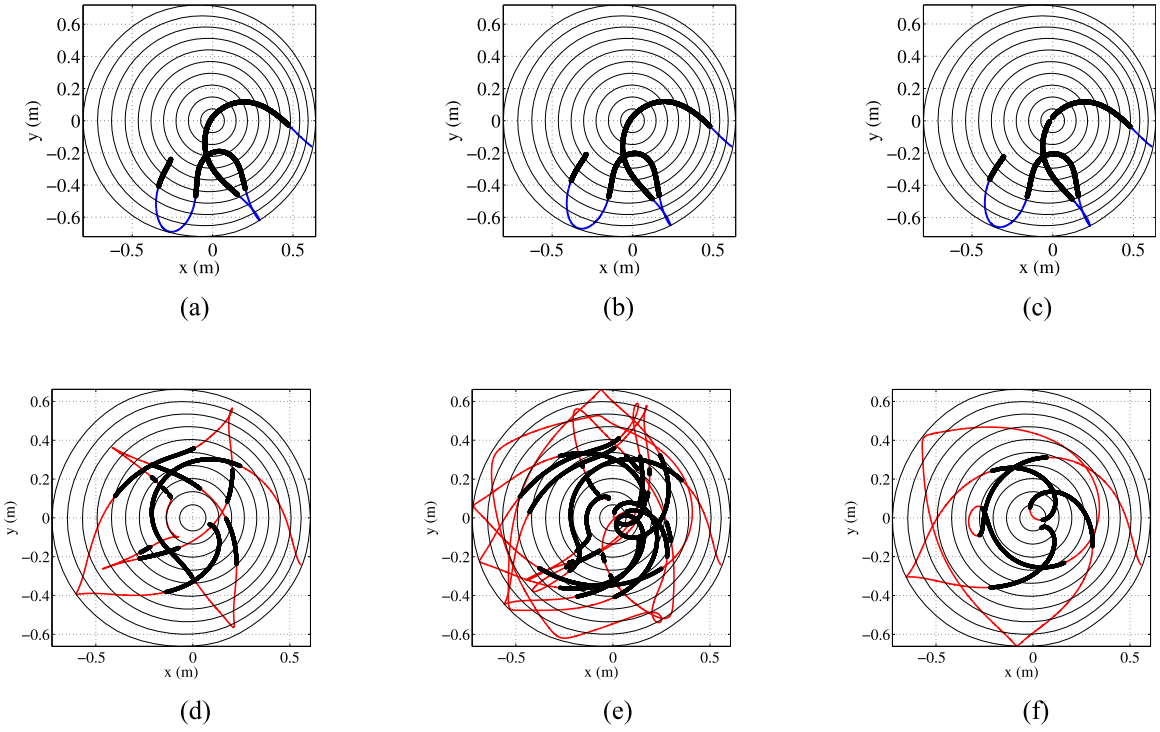


FIG. 3. Propagation of the LH ray corresponding to the main spectral lobe and the fourth waveguide row at three consecutive times for TS shot #45525: $t = 27$ s (a); $t = 28$ s (b); $t = 29$ s (c); and TS shot #45155: $t = 8.6$ s (d); $t = 8.8$ s (e); $t = 9.2$ s (f).

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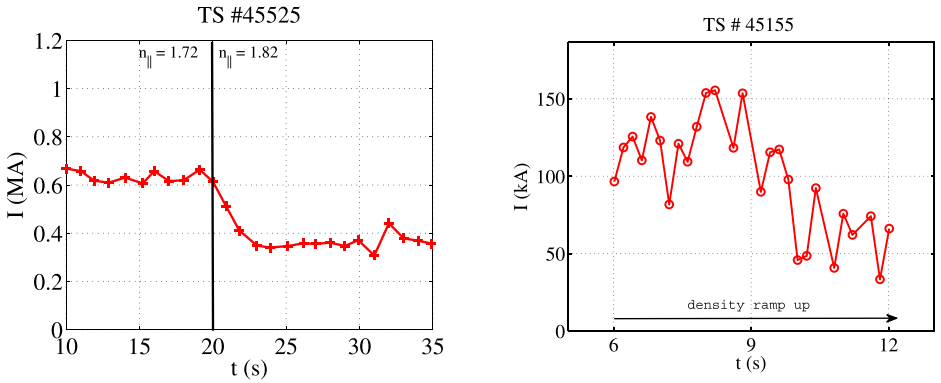


FIG. 4. LH driven current calculated by RTFP modelling. TS shot #45525 (a) illustrates a strong damping regime, while TS shot #45155 (b) illustrates a weak damping regime.

in the tail, which overlap at amplitude $1/e$, is adjusted to ensure spectral continuity. A fraction $\delta_P = P_{\text{tail}}/P_0$ of the initial power is transferred to the spectral tail, with either a flat or a linearly decreasing power profile in the tail. The grill and LCFS spectrum corresponding to a given lobe are illustrated in Fig. 6(a).

As the tail lobes are spectrally independent, they can be propagated using separate rays, each contributing to fill the spectral gap. In this process involving RTFP calculation, the self-consistency between the LH wave power flow and the electron distribution, as detailed in the Appendix, is essential. Parametric scans are conducted for a variety of LHCD discharges, and examples of both weak and strong damping cased are given in Figs. 6(b) and 6(c). In the weak damping case (TS #45155), it is found that a significant fraction δ_P of the power must be transferred to the spectral tail for the spectral gap to be properly bridged. In addition, the maximum value $n_{||\max}$ must reach a value close to $n_{||\text{LD}}$ evaluated at the center of the plasma. As expected, these conditions on $n_{||\max}$ and δ_P illustrate the fact that (1) the spectral tail must be wide enough to catch the Maxwellian distribution at the plasma center and (2) the electron distribution must be distorted sufficiently for significant damping to take place at $v_{||} \sim v_{||0}$. It is also found that five lobes are sufficient to describe the spectral tail and that using a flat or a linearly decreasing power profile in the tail has little effect on LHCD calculations. Remarkably, once the spectral gap is bridged, LHCD calculations do not seem to depend much upon variations of δ_P and $n_{||\max}$.

As the damping conditions in the plasma center are presumably independent of the mechanisms that could modify the spectrum at the edge, it is important to investigate the effect of power redistribution in the strong damping regime as well (TS #45525). Parametric scans illustrated in Figs.

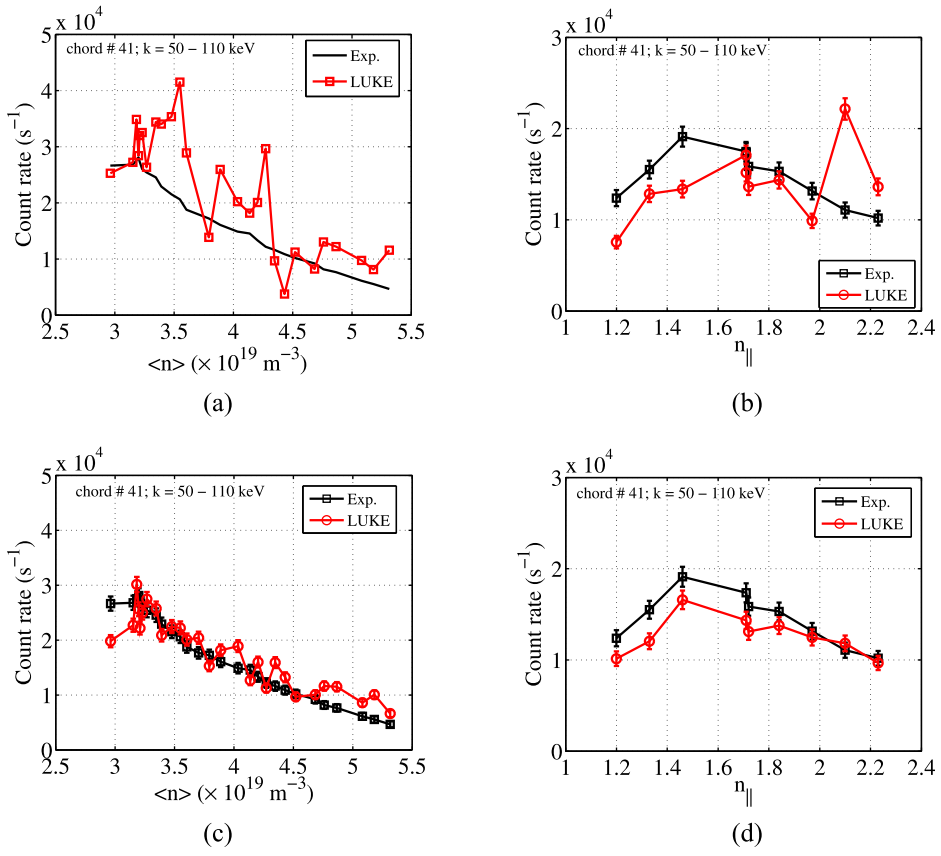


FIG. 5. Comparison between experimental HXR measurements and simulated reconstruction of the fast electron bremsstrahlung signal during LHCD for a density scan (TS shot #45155, left) and a $n_{||0}$ scan (TS shot #45519, right). Calculations in the top graphs (a) and (b) were obtained by the standard method, using the original spectrum to set initial conditions at the LCFS. Calculations in the bottom graphs (c) and (d) were obtained using the tail-model spectrum at the LCFS with $P_{\text{tail}}/P_0 = 0.5$ and $n_{||\max} = n_{||\text{LD}}$.

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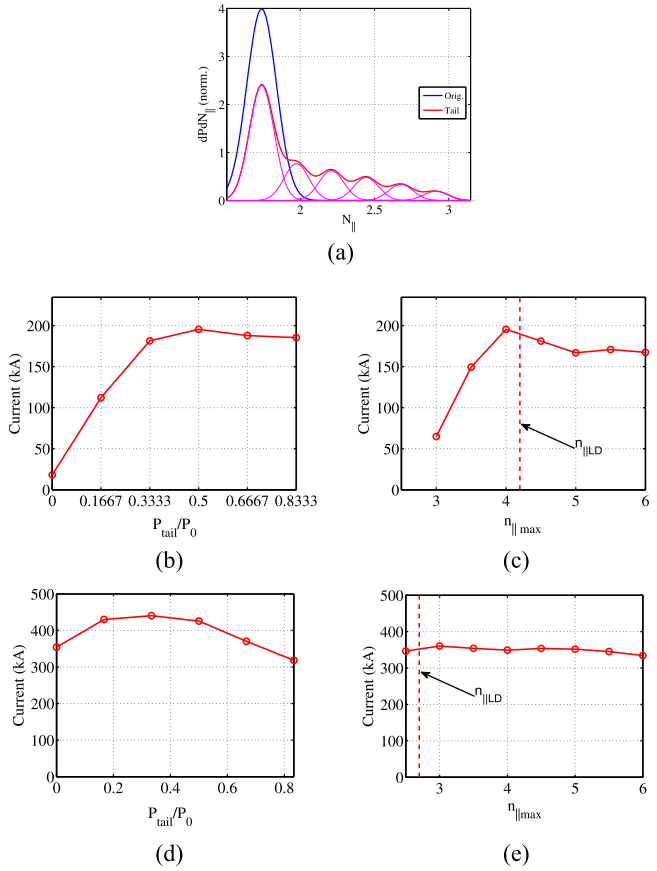


FIG. 6. Original Gaussian lobe describing the spectrum in front of the grill, compared to the spectrum at the LCFS constructed using a distribution of spectrally decoupled tail lobes (a). Evolution of the LH driven current as a function of the power fraction in the tail P_{tail}/P_0 (b) and the tail extension $n_{||\text{max}}$ (c) in the case of a weak damping regime as illustrated by TS shot 45155 at $t = 8.8$ s. Similar scans (d) and (e) are conducted for the case of a strong damping regime as illustrated by TS shot 45525 at $t = 27$ s.

6(b) and 6(c) show that power redistribution has little effect on the driven current, which remains at the level found without redistribution. This result is not surprising if we consider ray propagation and quasilinear dynamics as described in the Appendix and illustrated in Fig. 9(a). In the absence of *ab-initio* edge redistribution, a fraction of the power is redistributed instead by toroidal upshift, such that the resulting current is similar. Only in the weak damping regime, where toroidal upshift cannot bridge the spectral gap, can edge power redistribution play a significant role.

The robustness of the spectral tail model suggests to use it as a method for extending LHCD calculations to large spectral gap conditions.[26] The spectral tail is constructed systematically using five lobes with $\delta_P = 0.5$ and $n_{||\text{max}} = n_{||\text{LD}}$. This criterion for $n_{||\text{max}}$ is justified by the observation that LHCD calculations do not vary much with $n_{||\text{max}}$ once the spectral gap is bridged. This method also reduces to the standard RTFP calculation when the spectral gap vanishes such that $n_{||0} \geq n_{||\text{LD}}$. The spectral tail model is applied to LHCD parametric scans in Tore Supra.

The density is ramped up in TS shot #45155 where a large spectral gap is present (Fig. 4(b)). The HXR count rate measured experimentally along the central chord in the energy range of 50–110 keV, which is characteristic of LHCD fast electrons, is compared to the FEB reconstruction from LHCD simulations. While standard RTFP calculations show erratic results reflecting the stochastic behavior of the wave propagation, the results obtained using the spectral tail model agree remarkably well with experiments. No stochastic behavior is observed, a sign that the spectral gap is properly bridged such that the power is rapidly damped. In TS shot #45519, a $n_{||0}$ scan is provided by varying the phasing between waveguides. The experimental count rate displays a characteristic bell-shape dependence with a maximum near the accessibility limit ($n_{||a} = 1.4$). While the standard RTFP cannot account for this dependence, the spectral tail model again agrees well with experimental measurements.

While further comparisons—including other tokamaks—are needed to fully validate the spectral tail model, the excellent agreement shown for essential LHCD parametric dependencies in Tore Supra, combined with the robustness of calculations with respect to the spectral tail parametrization, allow the extension of LHCD calculations to the notoriously difficult weak damping regime. It also reinforces the assumption that the LH spectrum could be significantly modified at the plasma edge. The present model can actually be used to guide further research on the possible underlying mechanisms. For instance, it is compatible with the representation of spectra modified by PDI.[17]

IV. LHCD IN THE PRESENCE OF A FLUCTUATING SPECTRUM

One of the possible mechanisms suggested for bridging the spectral gap is the effect of density and magnetic field

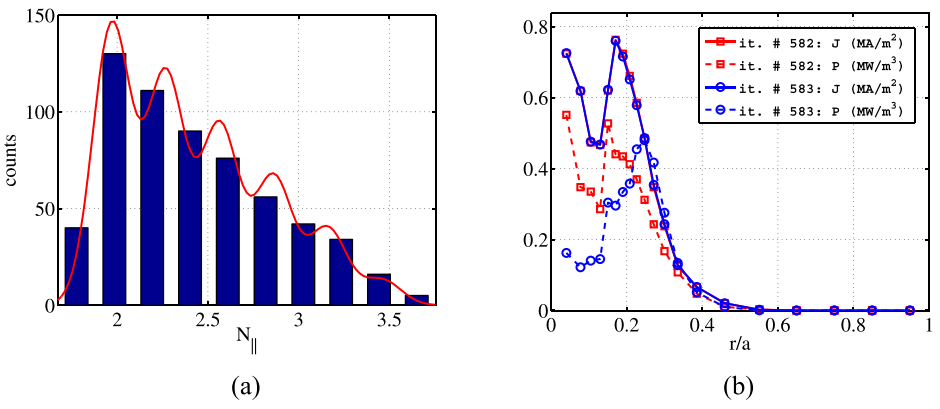


FIG. 7. (a) Distribution of $n_{||}$ over 600 fluctuations times (bars) based on the $n_{||}$ probability distribution from the spectral tail model in Fig. 6 (solid line); current density (solid lines) and power density (dotted lines) profiles at the consecutive fluctuation times #582 (squares) and #583 (circles).

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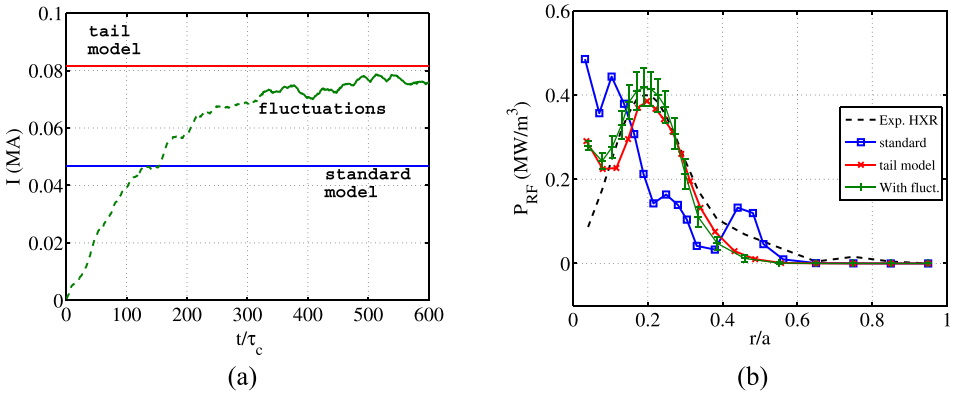


FIG. 8. (a) Evolution of LH driven current in the presence of a fluctuating spectrum, for TS shot #45519 with $n_{\|0} = 2$. Steady-state driven current obtained using the standard and tail-model calculations are shown for comparison. (b) Power deposition profile from standard and tail-model calculations, compared to the radial fast electron bremsstrahlung profile obtained by Abel inversion of the experimental HXR signal. The power deposition profile calculated with a fluctuating spectrum is averaged over the period $t/\tau_c = 315 - 600$ where the evolution has reached an asymptotic regime.

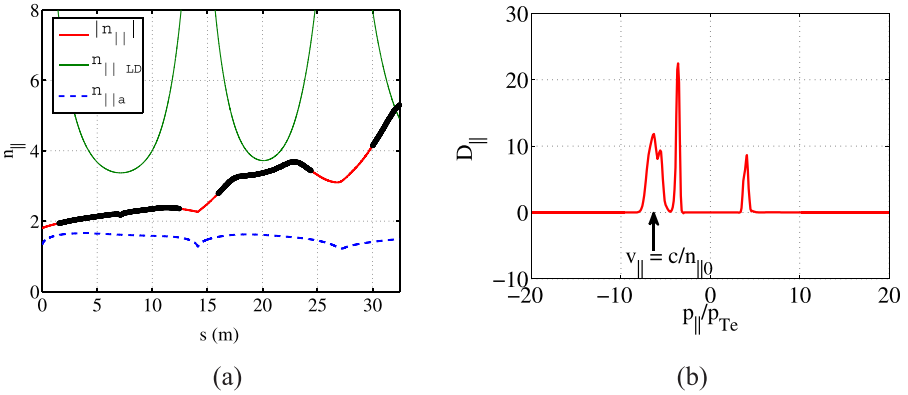


FIG. 9. (a) Evolution of $n_{\|}$ along the a ray trajectory for TS shot #45525. It is bounded by the accessibility condition $n_{\|a}$ and the strong linear damping condition $n_{\|\mathrm{LD}}$. Thick lines indicate sections of significant damping, which can occur far from the $n_{\|\mathrm{LD}}$ condition when quasilinear effects are important; (b) diffusion coefficient $D_{\|}$ as a function of $p_{\|}$, evaluated at the radial location of maximum power deposition and for $p_{\perp} = 0$. The resonant velocity corresponding to the injected $n_{\|0}$ is indicated.

fluctuations that could modify the LH spectrum in the scrape-off layer.[22–25] This effect can also be modelled using our spectral tail model.[26] Indeed, the LCFS spectrum shown in Fig. 6(a) can be interpreted as the probability distribution of a spectrum fluctuating between $n_{\|0}$ and $n_{\|\mathrm{max}}$. To demonstrate this equivalence, the evolution of LHCD is calculated in the presence of this fluctuating spectrum (see Fig. 7(a)) with a fluctuation time τ_f set to be equal to the thermal collision time τ_c.

While the power deposition profile varies significantly with each fluctuating state, as illustrated in Fig. 7(b), the driven current density profile does not vary much over one fluctuation time. This behavior results from the separation between the short fluctuation time τ_f of the spectrum and the long collision time $(v_{\|}/v_T)^3\tau_c$ of resonant electrons, which carry the current. Over one collision time, these electrons see an "average spectrum" and the spectral gap can be bridged dynamically as long as $\tau_f \ll (v_{\|}/v_T)^3\tau_c$. The total driven current increases steadily to an asymptotic regime close to the value obtained with the static spectrum (Fig. 8(a)).

As seen in Fig. 8(b), the power deposition profile averaged over the asymptotic regime is found to be very similar to the one obtained with the static spectrum model. These two equivalent models also agree very well with the HXR profile obtained by Abel inversion of experimental measurements. By contrast, the power deposition profile obtained with the standard method is much more central and in complete disagreement. This comparison contributes to validate further the spectral tail model.

V. CONCLUSION

Standard ray-tracing Fokker-Planck modelling fails in the presence of large spectral gaps that cannot be bridged by toroidal upshift. A new method based on a spectral tail created *ab initio* at the LCFS is used to extend LHCD to such configurations. The spectral tail model merges with standard calculations when the spectral gap vanishes. In the presence of large spectral gaps, essential LHCD parametric dependencies as provided by density and waveguide phasing scans are reproduced for the first time by simulation in the Tore Supra tokamak. The model can relate to the effects of PDI. It is also equivalent to LHCD calculations in the presence of a fluctuating spectrum, provided the fluctuation time is much shorter than the collision time of fast electrons. In this case, the modified spectrum can be interpreted as the time-average of a fluctuation spectrum. It compares very well with experimental measurements, which contributes to motivate further first-principle research on the effect of edge physics on LHCD.

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APPENDIX: SELF-CONSISTENT CALCULATION OF LH WAVE DAMPING AND THE ELECTRON DISTRIBUTION FUNCTION

The damping of RF waves on electrons in toroidal devices leads to a significant distortion of the distribution function, especially if the power is damped on suprathermal electrons as for LH waves. Conversely, the damping of RF waves through resonant wave-particle interaction depends on the details of the distribution function. Thus, RF wave damping and the particle distribution function must be calculated self-consistently.

Resonant interaction between electrons and finite-power RF waves is described within the quasilinear formalism, which is valid under the following conditions: first, the trapping time for the electron within the wavefield must be much longer than the interaction time between the electron and the wavepacket, such that non-linear trapping can be neglected; second, assuming that electrons follow periodic unperturbed orbits, the phase information must be lost between two consecutive passes through the wavepacket. The first condition, which sets a limit to the RF power, is equivalent to assuming a small incremental perturbation to the particle orbit. It makes it possible to describe wave-particle interaction as a diffusive process. The second condition assumes that additional perturbations of the particle orbit—due to collisions, turbulence, etc.—are sufficient to randomize the phase between consecutive passes. This condition may fail near low-n rational $q(\psi) = n/m$ flux-surface ψ, or when the interaction occurs within magnetic islands.

The electron distribution f results from the balance between the distorting effect of wave-particle interaction and the restoring effect of collisions towards a Maxwellian distribution. Its evolution is described by the bounce-averaged Fokker-Planck equation

$$\frac{\partial f}{\partial t} = C(f) + Q_{\mathrm{LH}}(f) + \cdots \tag{A1}$$

where $C(f)$ is the Fokker-Planck collision operator and $Q_{\mathrm{LH}}(f)$ is the quasilinear diffusion operator describing the interaction between electrons and LH waves. Assuming an axisymmetric toroidal configuration and a low-collisionality regime, Eq. (A1) is solved over the 3-D relativistic guiding-center phase-space where electrons orbits may be parameterized by (ψ, p, ξ), where ψ is the poloidal flux normalized to its value at the separatrix, p is the electron momentum, and $\xi = p_{||}/p$ is the pitch angle evaluated at the position of minimum magnetic field on the flux-surface ψ. Additional operators may describe ohmic heating, radial transport, ripple losses, etc. Within the zero-orbit width approximation, which is generally valid for the interaction between electrons and LH waves,[35] the quasilinear diffusion is directed along the parallel momentum and can be expressed as

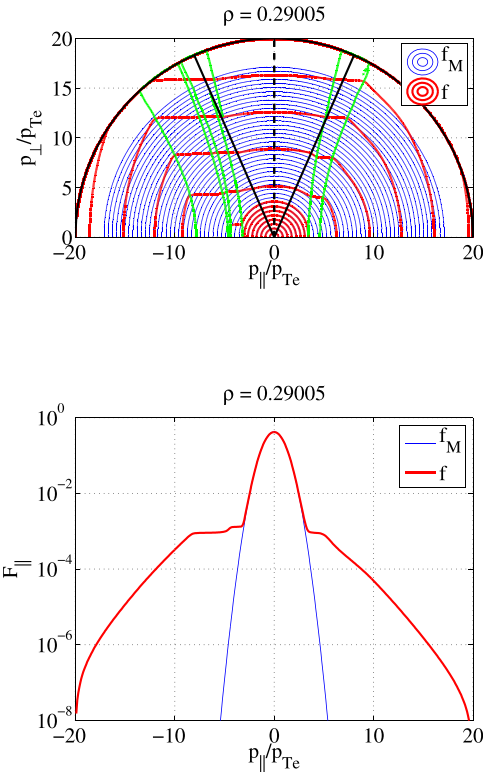


FIG. 10. Electron distribution function evaluated at the radial location of maximum power deposition $r/a = 0.29$, for TS shot #45525 at t = 27 s. The upper graph represents the 2-D distribution $f(p_{||}, p_{\perp})$, while the lower graph shows the corresponding 1-D distribution $F_{||}(p_{||}) = 2\pi \int_0^{\infty} f(p_{||}, p_{\perp}) p_{\perp} dp_{\perp}$.

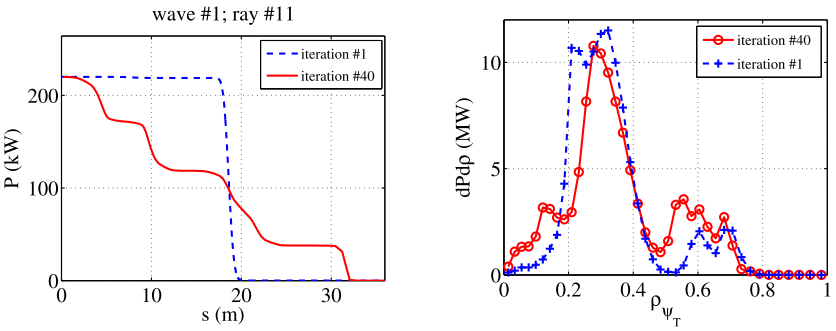
$$Q_{\mathrm{LH}}(f) = \frac{\partial}{\partial p_{||}} D_{||} \frac{\partial f}{\partial p_{||}}.$$

As stated in the introduction, the RTFP formalism assumes that the ordering $\lambda \ll d \ll L$ be satisfied, where λ is the wavelength, d is the beam size, and L is the equilibrium non-uniformity scalelength. The condition $\lambda \ll d$ is that of a well-collimated beam and ensures that beam characteristics such as the polarization and spectral properties and power flow are described by the ray propagation. The quasilinear diffusion coefficient includes the contributions of all rays crossing a given magnetic flux-surface ψ, including multiple passes of any given ray. It is decomposed as

$$D_{||}(\psi, p_{||}, p_{\perp}) = \sum_i \frac{\mathcal{P}_i}{\mathcal{P}_{i0}} \frac{D_{||i}}{\sqrt{\pi}\Delta n_{||i}} \exp\left[-\frac{(n_{||i} - n_{||\mathrm{res}})^2}{\Delta n_{||i}^2} \right], \tag{A2}$$

where $\mathcal{P}_i$ and $n_{||i}$ are the power and parallel index of refraction of the ray fragment i on the flux-surface ψ, respectively. The resonance condition yields $n_{||\mathrm{res}} = c/v_{||}$, where $v_{||}$ is the parallel velocity of the electron as it crosses the wavepacket. The diffusion coefficient $D_{||i}$ describes the intensity of the wave-particle interaction. Within the well-collimated beam approximation $\lambda \ll d$, a narrow spectral width $\Delta n_{||i} \sim c/(\omega d) \ll n_{||i}$ can be associated with a ray. It enters the calculation of the quasilinear diffusion operator (A2), where it provides a finite width to the resonance region in momentum space. As the evolution of $\Delta n_{||}$ along the ray propagation is not within reach

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(a) (b)

FIG. 11. Power flow along the ray: the linear estimate (#1) is compared to the quasilinear damping in the converged state (#40). The corresponding power deposition profiles are shown in graph (b).

of the RTFP formalism, and requires beam tracing instead, its initial value $\Delta n_{||i0}$ determined from the initial spectrum is maintained constant along the propagation.[36]

The condition $d \ll L$, which ensures that the beam sees a uniform plasma locally, is also necessary to build the quasilinear diffusion operator (A2). The electron guiding center parallel velocity, which varies along the trajectory due to the mirror force and enters the resonance condition $n_{||i} = c/v_{||}$, can be evaluated at the position where the ray crosses the flux-surface ψ. Without this condition, it would be necessary to construct the diffusion coefficient by numerical integration of the electron motion across the wavefield. The condition $d \ll L$ also means that wavefields corresponding to multiple rays or multiple passes on a given flux surface are less likely to overlap. In the presence of interfering wavepackets, the RTFP formalism fails unless the phase evolution is tracked along the rays. The multiplication of crossings due to the long ray propagation in the weak damping regime may lead to significant wavefield overlapping, which further invalidates RTFP formalism in this regime, unless edge mechanisms are invoked to shorten the rays as described in the present paper.

In the example provided by TS shot #45525 at t = 27 s (Fig. 1), rays representing the main lobe in the LH spectrum undergo three excursions toward the plasma center before the power is completely absorbed. The evolution of $n_{||}$ along the ray trajectory (Fig. 9(a)) shows a succession of upshifts that eventually bridge the spectral gap. Their respective contributions add up to form a broad $D_{||}$ profile, as seen in Fig. 9(b), which fills the spectral gap between the Maxwellian distribution and the injected $n_{||0}$. As shown in Fig. 10, the resulting electron distribution exhibits a wide suprathermal tail, which is clearly related to the broad $D_{||}$ profile and demonstrates that the spectral gap is filled.

The density of power deposited by LH waves on the flux-surface ψ is a moment of the electron distribution given by

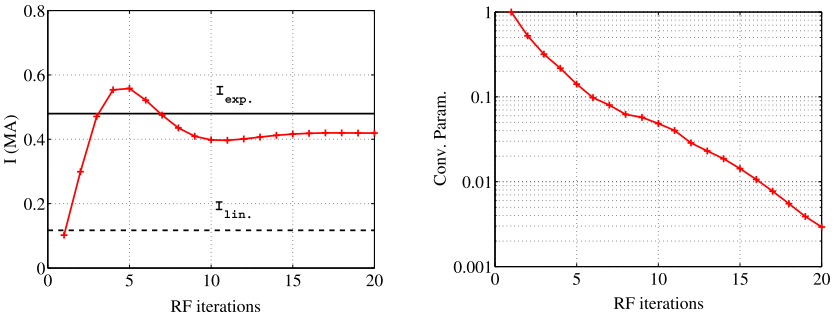
$$P_{\rm abs}(\psi) = \langle (\gamma - 1) mc^2 Q_{\rm LH}(f) \rangle,$$

where $\langle \mathcal{O}(f) \rangle$ represent the flux-surface averaged moment of the operator $\mathcal{O}(f)$ and $\gamma = \sqrt{1 + p^2/(mc)^2}$ is the relativistic factor. $P_{\rm abs}(\psi)$ can also be decomposed into the contributions from each ray fragment i as $P_{\rm abs} = \sum_i P^i_{\rm abs}$. In the $\Delta n_{||i} \to 0$ limit, $P^i_{\rm abs}$ reduces to

$$P^i_{\rm abs}(\psi) = D_{||i} \frac{\mathcal{P}_i}{\mathcal{P}_{i0}} \frac{c}{n_{||i}} \left\langle \frac{\partial f}{\partial p_{||}} \bigg|_{v_{||} = c/n_{||i}} \right\rangle.$$

The power deposited by each ray fragment i on the flux-surface ψ depends upon the derivative of the distribution function in the parallel momentum direction, and reduces to the linear limit for a Maxwellian distribution $f = f_M$. For the high $n_{||}$ components, the quasilinear absorption tends to be lower than the linear limit because the distribution is flattened in the parallel direction. For the low $n_{||}$ components, the quasilinear absorption is much higher than the linear limit because a tail is pulled in the distribution function such that $f \gg f_M$ for the corresponding resonant $v_{||} = c/n_{||i}$.

The difference between linear and quasilinear absorption is illustrated by plotting the power carried by a given ray along its trajectory (Fig. 11(a)). In the first iteration (#1) of our convergence scheme, linear damping is assumed as an initial guess. The iteration #40 corresponds to a converged state with self-consistent calculation of the LH damping with the distribution function. The lower $n_{||}$ components are now absorbed in the presence of the quasilinear tail if the electron distribution. Remarkably, the linear and quasilinear power



(a) (b)

FIG. 12. (a) Evolution of the driven current in the iterative scheme between f and $D_{||}$. The convergence parameter (b) is calculated between two consecutive iteration states $k-1$ and k. It accounts for all ray fragments i and is defined as $\max_i[|\mathcal{P}^k_i - \mathcal{P}^{k-1}_i|/\mathcal{P}_{i0}]$.

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deposition profiles are very similar (Fig. 11(b)). This is not surprising as quasilinear damping occurs where a fast electron tail is formed, which is precisely where linear damping is expected. Consequently, the power and current deposition profiles are poor indicators of the convergence in the self-consistent calculation LH damping with the electron distribution. It is essential to measure the convergence on the power flow along each ray instead. The evolution of the driven current during the iterative process is shown in Fig. 12, where a convergence parameter is also defined. It shows that the driven current is very sensitive to details of the quasilinear plateau build-up, but eventually converges towards a value very close to experimental measurements.

[1]N. J. Fisch, Phys. Rev. Lett. **41**, 873 (1978).
[2]E. Nilsson, J. Decker, Y. Peysson, J.-F. Artaud, A. Ekedahl, J. Hillairet, T. Aniel, V. Basiuk, M. Goniche, F. Imbeaux, D. Mazon, and P. Sharma, Nucl. Fusion **53**, 083018 (2013).
[3]H. Takahashi, Phys. Plasmas **1**, 2254 (1994).
[4]Y. F. Baranov and V. I. Fedorov, Sov. Tech. Phys. Lett. **4**, 322 (1978).
[5]M. Brambilla and A. Cardinali, Plasma Phys. **24**, 1187 (1982).
[6]P. Bonoli and R. Englade, Phys. Fluids **29**, 2937 (1986).
[7]R. Harvey and M. McCoy, "The CQL3D Fokker-Planck code," in *IAEA Technical Committee on Advances in Simulation and Modeling of Thermonuclear Plasmas* (1992), pp. 489–526.
[8]Y. Peysson and J. Decker, AIP Conf. Proc. **1069**, 176 (2008).
[9]M. Goniche, V. Basiuk, J. Decker, P. K. Sharma, G. Antar, G. Berger-By, F. Clairet, L. Delpech, A. Ekedahl, J. Gunn, J. Hillairet, X. Litaudon, D. Mazon, E. Nilsson, T. Oosako, Y. Peysson, M. Preynas, M. Prou, J. L. Segui *et al.*, Nucl. Fusion **53**, 033010 (2013).
[10]A. Schmidt, P. T. Bonoli, O. Meneghini, R. R. Parker, M. Porkolab, S. Shiraiwa, G. Wallace, J. C. Wright, R. W. Harvey, and J. R. Wilson, Phys. Plasmas **18**, 056122 (2011).
[11]J. Decker, E. Nilsson, Y. Peysson, J.-F. Artaud, A. Ekedahl, J. Hillairet, T. Aniel, M. Goniche, T. Hoang, F. Imbeaux, D. Mazon, and P. K. Sharma, "Modelling of LHCD at various densities in Tore Supra tokamak," in 39th EPS Conference on Plasma Physics and Controlled Fusion, 2012.
[12]J. P. Bizarro and D. Moreau, Phys. Fluids B **5**, 1227 (1993).
[13]J. C. Wright, P. T. Bonoli, A. E. Schmidt, C. K. Phillips, E. J. Valeo, R. W. Harvey, and M. A. Brambilla, Phys. Plasmas **16**, 072502 (2009).
[14]S. Shiraiwa, O. Meneghini, R. Parker, P. Bonoli, M. Garrett, M. C. Kaufman, J. C. Wright, and S. Wukitch, Phys. Plasmas **17**, 056119 (2010).
[15]S. Moriyama, Y. Nakamura, A. Nagao, E. Jotaki, K. Nakamura, N. Hiraki, and S. Itoh, Nucl. Fusion **30**, 47 (1990).
[16]Y. Peysson, E. Sebelin, X. Litaudon, D. Moreau, J. C. Miellou, M. Shoucri, and I. P. Shkarofsky, Nucl. Fusion **38**, 939 (1998).
[17]R. Cesario, A. Cardinali, C. Castaldo, F. Paoletti, and D. Mazon, Phys. Rev. Lett. **92**, 175002 (2004).
[18]R. Cesario, A. Cardinali, C. Castaldo, F. Paoletti, W. Fundamenski, S. Hacquin, and the JET-EFDA Workprogramme Contributors, Nucl. Fusion **46**, 462 (2006).
[19]S. G. Baek, R. R. Parker, S. Shiraiwa, G. M. Wallace, P. T. Bonoli, D. Brunner, I. C. Faust, A. E. Hubbard, B. LaBombard, and M. Porkolab, Plasma Phys. Controlled Fusion **55**, 052001 (2013).
[20]M. Preynas, M. Goniche, J. Hillairet, X. Litaudon, A. Ekedahl, and L. Colas, Nucl. Fusion **53**, 013012 (2013).
[21]G. M. Wallace, A. E. Hubbard, P. T. Bonoli, I. C. Faust, R. W. Harvey, J. W. Hughes, B. L. LaBombard, O. Meneghini, R. R. Parker, A. E. Schmidt, S. Shiraiwa, A. P. Smirnov, D. G. Whyte, J. R. Wilson, J. C. Wright, S. J. Wukitch, and the Alcator C-Mod Team, Nucl. Fusion **51**, 083032 (2011).
[22]P. Bonoli and E. Ott, Phys. Rev. Lett. **46**, 424 (1981).
[23]N. Bertelli, G. Wallace, P. T. Bonoli, R. W. Harvey, A. P. Smirnov, S. G. Baek, R. R. Parker, C. K. Phillips, E. J. Valeo, J. R. Wilson, and J. C. Wright, Plasma Phys. Controlled Fusion **55**, 074003 (2013).
[24]Y. Peysson, J. Decker, L. Morini, and S. Coda, Plasma Phys. Controlled Fusion **53**, 124028 (2011).
[25]K. Hizanidis, A. Ram, Y. Kominis, and C. Tsironis, Phys. Plasmas **17**, 1 (2010).
[26]J. Decker, Y. Peysson, J.-F. Artaud, E. Nilsson, A. Ekedahl, M. Goniche, J. Hillairet, and D. Mazon, AIP Conf. Proc. **1580**, 129 (2014).
[27]J. F. Artaud, V. Basiuk, F. Imbeaux, M. Schneider, J. Garcia, G. Giruzzi, P. Huynh, T. Aniel, F. Albajar, J. M. Ane, A. Becoulet, C. Bourdelle, A. Casati, L. Colas, J. Decker, R. Dumont, L. G. Eriksson, X. Garbet, R. Guirlet, P. Hertout, G. T. Hoang, W. Houlberg, G. Huysmans, E. Joffrin, S. H. Kim, F. Kochl, J. Lister, X. Litaudon, P. Maget, R. Masset, B. Pegourie, Y. Peysson, P. Thomas, E. Tsitrone, and F. Turco, Nucl. Fusion **50**, 043001 (2010).
[28]Y. Peysson, J. Decker, and L. Morini, Plasma Phys. Controlled Fusion **54**, 045003 (2012).
[29]J. Hillairet, D. Voyer, A. Ekedahl, M. Goniche, M. Kazda, O. Meneghini, D. Milanesio, and M. Preynas, Nucl. Fusion **50**, 125010 (2010).
[30]J. Decker and Y. Peysson, Report EUR-CEA-FC-1736, Euratom-CEA (2004).
[31]J. Decker and Y. Peysson, "On self-consistent simulation of the lower hybrid current drive," in 33rd EPS Conference on Plasma Physics and Controlled Fusion, 2006.
[32]Y. Peysson and J. Decker, Phys. Plasmas **15**, 092509 (2008).
[33]R. Cesario, L. Amicucci, A. Cardinali, C. Castaldo, M. Marinucci, L. Panaccione, F. Santini, O. Tudisco, M. L. Apicella, G. Calabro, C. Cianfarani, D. Frigione, A. Galli, G. Mazzitelli, C. Mazzotta, V. Pericoli, G. Schettiniv, A. A. Tuccillo, and the FTU Team, Nat. Commun. **1**, 55 (2010).
[34]B. J. Ding, E. H. Kong, M. H. Li, L. Zhang, W. Wei, M. Wang, H. D. Xu, Y. C. Li, B. L. Ling, Q. Zang, G. S. Xu, X. F. Han, H. L. Zhao, L. Zhang, L. M. Zhao, H. C. Hu, Y. Yang, L. Liu, A. Ekedahl, M. Goniche, R. Cesario, Y. Peysson, J. Decker, V. Basiuk, P. Huynh, J. Artaud, F. Imbeaux, J. F. Shan, F. K. Liu, Y. P. Zhao, X. Z. Gong, L. Q. Hu, X. Gao, H. Y. Guo, B. N. Wan, J. G. Li, and the EAST Team, Nucl. Fusion **53**, 113027 (2013).
[35]J. Decker, Y. Peysson, A. J. Brizard, and F.-X. Duthoit, Phys. Plasmas **17**, 112513 (2010).

[36]Depending upon the (ψ, p, ξ) grid, it is sometimes necessary to increase $\Delta n_{\parallel i}$ for numerical application in order to calculate (A2) with acceptable accuracy. Above this minimum value imposed by the grid step size, $\Delta n_{\parallel}$ variations are found to have little effect on the wave damping and electron distribution.

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